An Efficient Method of License Plate Location

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***Abstract*—This paper presents a novel approach of license plate location. The proposed algorithm involves the following three steps. First, the vertical edges of the vehicle image are extracted by Sobel operator. Second, HSV color space and integral image are employed to locate candidates in yellow license plates and non-yellow license plates. Finally, connected component analysis is to locate the region of license plate accurately. Experimental results on a large volume of natural-scene vehicle plate image sets, which are extracted from low-quality video sequences, demonstrate that our technique achieves a verification rate of around 95% on yellow license plates and 99% on non-yellow ones. The total time of processing one yellow image is less than 0.1s and the non-yellow one is less than 0.05s, meeting the requirements of real-time application.**

I. Introduction

With the rapid development of Intelligent Transportation Systems (ITS), License Plate Recognition (LPR) has been playing an important role in people's life during the past few years, such as electronic payment systems (toll payment, parking fee payment), freeway, and arterial management systems for surveillance [1].

As far as the location of the license plate(LP) region is concerned, a lot of methods were reported in previous research. The most prevalent technique is the methods based on edge statistics, which are based on the principle that the change of brightness in the LP region is more remarkable and more frequent than elsewhere. But hardly can they be applied to complex images because of their sensitivity to the unwanted edges with high magnitude or variance, such as the radiator, the dazzle lamp, and characters on the vehicle. [2], [3], [4], [5] combined edge statistics with morphological steps to eliminate unwanted edges in the processed images.

Another important technique is color-based methods, which make use of the color combination of a plate (background) and characters (foreground). This combination is unique and occurs almost only in the plate region [6], [7], [8], [9]. However, the solutions do not provide a high degree of accuracy in natural scenery, since the vast majority of color combinations are not stable when the lighting conditions change, such as the blue-white combination.

Genetic algorithms [10], Hough transform [11], and wavelet transform [12] are also incorporated for LP detection.

This paper further researches the subject of license plate location. As the rectangle of license plate contains rich edges and texture information, we consider it based on edge information but different to the previous ones. In this paper, we first extract the vertical edge image by Sobel operator, and then use the HSV color model and integral image to locate the candidate, and finally use connected component analysis to locate the license plate accurately.

This paper is organized as follows. Sec.II introduces the key techniques in our approach. Sec.III describes the steps of LP location in details. Then, Experimental results are detailedly shown in Sec.IV, followed by conclusion in Sec.V.

II. The key techniques in our framework

A. HSV color space

The HSV (hue, saturation, intensity) system is a extensively used color space in image processing, which is compatible with human vision. Inspired by the fact that the hue of yellow is less insensitive to lighting condition than the other colors(such as the blue), we make use of HSV color space to locate the yellow license plates.

The conversion from RGB color space to HSV color space is [13]:

$$\begin{cases} V = max(R,G,B) \\ S = \begin{cases} [V - min(R,G,B)] \times 255/V & V \neq 0 \\ 0 & \text{otherwise} \end{cases} \\ H = \begin{cases} (G-B) \times 60/S & V = R \\ 180 + (B-R) \times 60/S & V = G \\ 240 + (R-G) \times 60/S & V = B \end{cases} \\ H = H + 360 \quad if H < 0 \end{cases} \tag{1}$$

where R, G, and B denote the red, green, and blue values in RGB color space. H, S, and V denote the hue, saturation, and intensity values in HSV color space.

B. Integral image

An integral image is an algorithm for generating the sum of values in a rectangular subset of a grid quickly and efficiently. It was first introduced to the computer graphics world in 1984 [14] and was prominent used in the Viola and Jones' robust real-time object detection framework [15].

As shown as in Fig. 1, the value at any point (x, y) in the summed area table, i.e. $sat(x, y)$, is just the sum of all the pixels above and to the left of (x,y):

$$sat(x,y) = \sum_{x' \leq x, y' \leq y} i(x', y') \tag{2}$$

where $i(x', y')$ denotes the gray-level value at the point (x', y').

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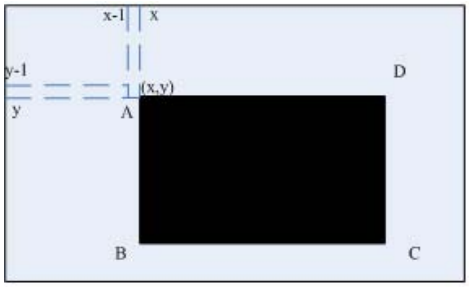

Fig. 1. Integral image

Moreover, the summed area table can be computed efficiently in a single pass over the image, using the fact that the value in the summed area table at (x,y) is just:

$$\begin{aligned} sat(x,y) &= sat(x-1,y) + s(x,y) \\ s(x,y) &= s(x,y-1) + i(x,y) \end{aligned} \tag{3}$$

where $s(x,y)$ denotes the sum of all the pixels above the point (x,y) at the column y. And $s(x,0)=0$, $sat(0,y)=0$.

Once the summed area table has been computed, the task of evaluating any rectangle can be accomplished in constant time with just four array references:

$$\sum_{regionABCD} i(x',y') = sat(A) + sat(C) - sat(B) - sat(D) \tag{4}$$

III. The Proposed Algorithm for License Plate Location

There are two facts that attract our attention: one is that the texture information of LP region is different from the other regions in the edge-detected image, such as the edge numbers in a specific window and in the horizontal direction; the other is that the hue of the yellow is insensitive to lighting condition change. Thus we locate the license plate by the combination of edge, texture and color information.

Fig. 2 depicts the flowchart of the whole system. Firstly, edge-detected image is obtained by vertical Sobel detector. Secondly, HSV color space is exploited to yellow license plates, and texture analysis is operated in non-yellow license plates. Finally, connected component analysis is used to locate the license plate.

A. Vertical edge extraction

We select the vertical Sobel operator to detect the vertical edges, because the plate area contains rich edges information and the computational time of Sobel operator is low. After convolving the vehicle image with the vertical Sobel operator to get the vertical gradient image, we compute the mean of the absolute gradient values in the vertical gradient image and multiply it by a coefficient as a threshold (for example 4Grad), or compute the gradient histogram and find a gradient at a certain percentage (for example 75%) of the gradient distribution as a threshold. Then the binary vertical edge image(BE) is obtained using this threshold.

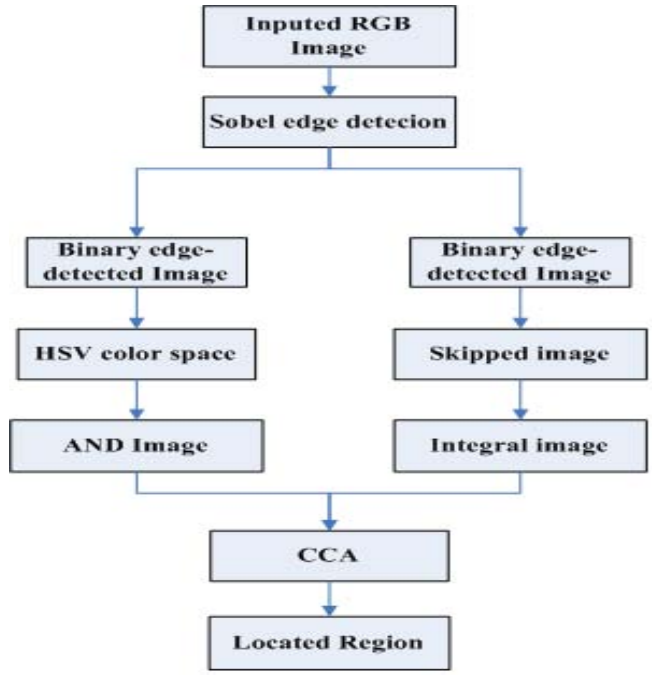

Fig. 2. Flowchart of the proposed approach

B. Yellow license plates location

In contrast to non-yellow license plates, especially blue license plates which are sensitive to lighting conditions change, yellow license plates are stable. For example, some blue license plates are like black license plates at night.

In the HSV color space, the hue of yellow is according to the $60°$ in the HSV color wheel. The yellow regions are obtained as following:

$$BH(i,j) = \begin{cases} 0 & H(i,j) \geq 75° \mid H(i,j) \leq 45° \\ 1 & \text{otherwise} \end{cases} \tag{5}$$

where $H(i,j)$ is the hue of pixel in (i,j). $BH(i,j)$ is the binary image in which the yellow regions are white and the non-yellow regions are black.

As shown as in Fig. 3, in order to eliminate the edge-rich regions in the background (such as the radiator and dazzle lamp region), the binary edge image(BE) and the binary hue image(BH) are operated by logical AND.

C. Operation on non-yellow license plate

As the texture of LP is different from the one of non-license plates regions, we make use of sum of edge and skip quantity to locate candidates in non-yellow license plates, where sum of edge is the quantity of pixels in a rectangle of BE, and the skip quantity is the skip number from white pixel to black pixel or from black pixel to white pixel in the rectangle. Calculating the sum of edge is similar to the integral image. And the skip quantity ($S(x,y)$) can be obtained as follows:

$$S(x,y) = S'(x+w/2,y) - S'(x-w/2,y) \tag{6}$$

where

$$S'(x,y) = S'(x-1,y) + \overline{\delta_{(x-1,y),(x,y)}}$$
$$\overline{\delta_{(x-1,y),(x,y)}} = \begin{cases} 1 & BE(x-1,y) \neq BE(x,y) \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

Fig. 4 shows an example of non-yellow license plates location.

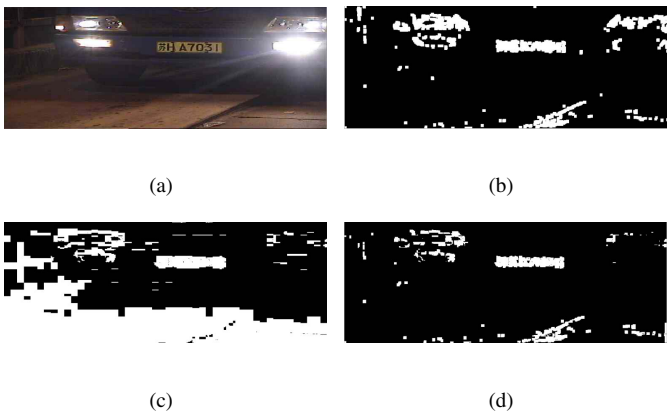

Fig. 3. Yellow license plate location. (a) Initial RGB image, (b) Sobel-dilated image, (c) binary hue image, (d) Image after (b) AND (c)

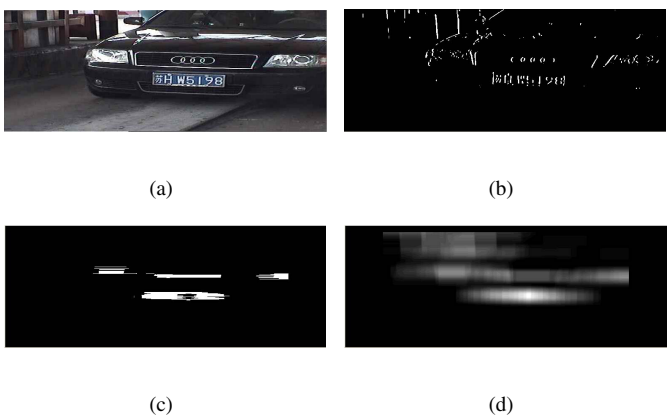

Fig. 4. Non-Yellow license plate location. (a) Initial RGB image, (b) Binary edge detection image, (c) Skip image, (d) Integral image

D. Connected component analysis

After the operations in Sec.III-B and Sec.III-C, there are a few candidates. The technique of connected component analysis(CCA) is exploited to locate the license plate region accurately.

1) Combining close regions: The candidate regions are combined if they are almost the same y-direction position and there are small distance between them. It can be formulated by the following equation:

$$\begin{cases} dis_w < T_1 \times \sqrt{w_1 \cdot w_2} \\ \frac{Overlap_h}{min\{h_1,h_2\}} > T_2 \end{cases} \quad (8)$$

where the $Overlap_h$ denote the overlapped length of two rectangles in vertical direction, and dis_wdenote the distance of horizontal direction, and the w_1, w_2, h_1, h_2denote the widths and heights of rectangles respectively. T_1 and T_2 are the thresholds, which are set to 0.5 and 0.6 in our paper. Two rectangles are combined if the equation holds.

2) membership of regions: With the important information from the CCA, we get the memberships of some features in different regions, such as the membership of aspect ratio(AR), area(A), height(H), width(W), and extent(E,computed as the Area divided by the area of the bounding box). Fig. 5 shows these memberships. Then the weighted membership is obtained as following:

$$memship = 0.4AR + 0.15A + 0.15W + 0.15H + 0.15E \quad (9)$$

LP region is the candidate with the biggest membership.

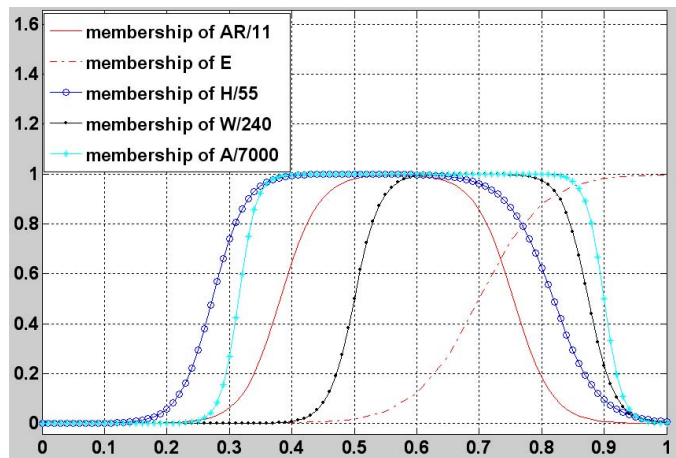

Fig. 5. The memberships of features.

IV. Experiments

In order to evaluate the proposed approach, a test set of 2943 license plate images from the real world were processed, which were taken from low-quality surveillance video at Shanghai-Nanjing Expressway Toll gate in China. And the images consisted of 768×288 pixels. A set of 2847 images from the test was correctly located. Performance is carried out in $C++$ on computer with Pentium-4 2.4GHz, 1GB RAM. As shown as in TableI, the proposed algorithm detect the region of the yellow license plate above 95% and the non-yellow one above 98%, which implies that our proposed system is robust. The reasons that the location rate of the non-yellow license plate is higher than the yellow one are that there maybe dirt and physical damage in yellow license plate vehicles. The average processing time of one yellow color image and one non-yellow color image are about $92ms$ and $47ms$ respectively, which means our technique meets the requirements of real-time processing. Fig. 6 shows the detection result of some images.

V. Conclusion

This paper presents a simple license plate detection algorithm from license plate images in a novel perspective to

TABLE I
THE RESULTS OF EXPERIMENTS(YP:YELLOW PLATES, NYP:NON-YELLOW PATES)

	Quantity	Location success	success rate	average time
YP	1908	1823	95.55%	$92ms$
NYP	1035	1024	98.94%	$47ms$

generate more reliable and accurate location under ambiguity effectively and efficiently. The proposed system can be divided into three key sections, that is, Sobel edge detection, HSV color space and integral image, and connected component analysis. A test with a set of 2943 real plate images was processed during the experiments. Results show that our license plate detection algorithm can achieve both good precision rate and low time complexity. Thus, this novel approach can be considered as a promising alternative technique for license plate detection. In the future, we focus on the vehicle license plates location in multi-view .

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(a)

(b)

(c)

(d)

Fig. 6. Segmentation results from degraded license plate images. (a and b) examples of yellow license plates location, (c and d) examples of non-yellow license plates location,

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